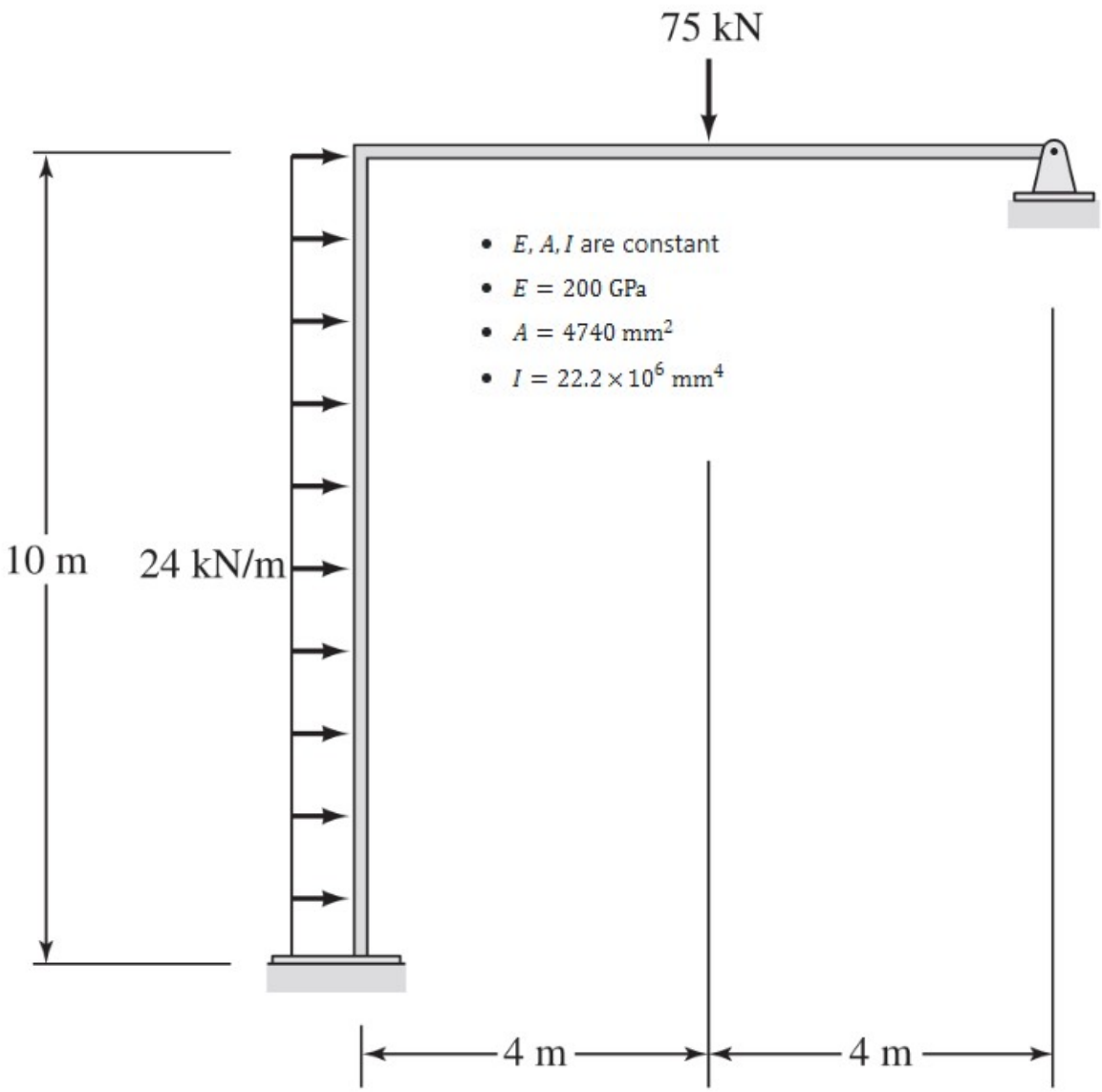


$E := 200\ GPa$ $\quad A := 4740\ mm^2$ $\quad I := 22.2 \cdot 10^6\ mm^4 = 0.0000222\ m^4$

$L_1 := 10\ m$ $\quad L_2 := 8\ m$ $\quad x_1 := 0\ m$ $\quad x_2 := 0\ m$ $\quad x_3 := 8\ m$

$A = (4.74 \cdot 10^{-3})\ m^2$ $\quad y_1 := 0\ m$ $\quad y_2 := 10\ m$ $\quad y_3 := 10\ m$

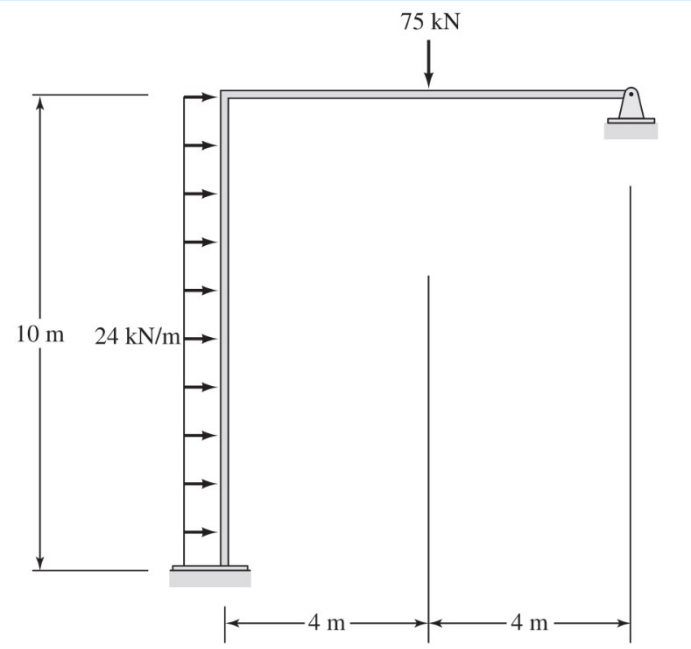
For this question, have to do this unitless on mathcad because mathcad struggles to combine these units in multiplication, keeping units in terms of kN and m

$E := 200000000$ kN/m^2 $\quad A := 4.74 \cdot 10^{-3}$ m^2 $\quad I := 0.0000222$ m^4

$L_1 := 10$ $\quad L_2 := 8$ $\quad x_1 := 0$ $\quad x_2 := 0$ $\quad x_3 := 8$ $\quad y_1 := 0$ $\quad y_2 := 10$ $\quad y_3 := 10$

.com for more information.

Marking the joints and DOFs



$u_{x1} = fixed$ $\quad u_{x2} = free$ $\quad u_{x3} = fixed$

$u_{y1} = fixed$ $\quad u_{y2} = free$ $\quad u_{y3} = fixed$

$\theta_1 = fixed$ $\quad \theta_2 = free$ $\quad \theta_3 = free$

$$k_{frame} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & \frac{-AL^2}{I} & 0 & 0 \\ 0 & 12 & 6\,L & 0 & -12 & 6\,L \\ 0 & 6\,L & 4\,L^2 & 0 & -6\,L & 2\,L^2 \\ \frac{-AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & 0 \\ 0 & -12 & -6\,L & 0 & 12 & -6\,L \\ 0 & 6\,L & 2\,L^2 & 0 & -6\,L & 4\,L^2 \end{bmatrix}$$

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See www.mathcad.com for more information.

Element One:

$$k_1 := \frac{E \cdot I}{L_1^{\,3}} \begin{bmatrix} \frac{A \cdot L_1^{\,2}}{I} & 0 & 0 & \frac{-A \cdot L_1^{\,2}}{I} & 0 & 0 \\ 0 & 12 & 6 \cdot L_1 & 0 & -12 & 6 \cdot L_1 \\ 0 & 6 \cdot L_1 & 4 \cdot L_1^{\,2} & 0 & -6 \cdot L_1 & 2 \cdot L_1^{\,2} \\ \frac{-A \cdot L_1^{\,2}}{I} & 0 & 0 & \frac{A \cdot L_1^{\,2}}{I} & 0 & 0 \\ 0 & -12 & -6 \cdot L_1 & 0 & 12 & -6 \cdot L_1 \\ 0 & 6 \cdot L_1 & 2 \cdot L_1^{\,2} & 0 & -6 \cdot L_1 & 4 \cdot L_1^{\,2} \end{bmatrix}$$

$$k_1 = \begin{bmatrix} 9.48 \cdot 10^4 & 0 & 0 & -9.48 \cdot 10^4 & 0 & 0 \\ 0 & 53.28 & 266.4 & 0 & -53.28 & 266.4 \\ 0 & 266.4 & 1.776 \cdot 10^3 & 0 & -266.4 & 888 \\ -9.48 \cdot 10^4 & 0 & 0 & 9.48 \cdot 10^4 & 0 & 0 \\ 0 & -53.28 & -266.4 & 0 & 53.28 & -266.4 \\ 0 & 266.4 & 888 & 0 & -266.4 & 1.776 \cdot 10^3 \end{bmatrix}$$

$$c_1 := \frac{x_2 - x_1}{L_1} = 0 \qquad s_1 := \frac{y_2 - y_1}{L_1} = 1$$

$$T_1 := \begin{bmatrix} c_1 & s_1 & 0 & 0 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_1 & s_1 & 0 \\ 0 & 0 & 0 & -s_1 & c_1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$K_1 := T_1^{\,\mathrm{T}} \cdot k_1 \cdot T_1 = \begin{bmatrix} 53.28 & 0 & -266.4 & -53.28 & 0 & -266.4 \\ 0 & 9.48 \cdot 10^4 & 0 & 0 & -9.48 \cdot 10^4 & 0 \\ -266.4 & 0 & 1.776 \cdot 10^3 & 266.4 & 0 & 888 \\ -53.28 & 0 & 266.4 & 53.28 & 0 & 266.4 \\ 0 & -9.48 \cdot 10^4 & 0 & 0 & 9.48 \cdot 10^4 & 0 \\ -266.4 & 0 & 888 & 266.4 & 0 & 1.776 \cdot 10^3 \end{bmatrix}$$

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Element Two:

$$k_2 := \frac{E \cdot I}{L_2^{\,3}} \begin{bmatrix} \frac{A \cdot L_2^{\,2}}{I} & 0 & 0 & \frac{-A \cdot L_2^{\,2}}{I} & 0 & 0 \\ 0 & 12 & 6 \cdot L_2 & 0 & -12 & 6 \cdot L_2 \\ 0 & 6 \cdot L_2 & 4 \cdot L_2^{\,2} & 0 & -6 \cdot L_2 & 2 \cdot L_2^{\,2} \\ \frac{-A \cdot L_2^{\,2}}{I} & 0 & 0 & \frac{A \cdot L_2^{\,2}}{I} & 0 & 0 \\ 0 & -12 & -6 \cdot L_2 & 0 & 12 & -6 \cdot L_2 \\ 0 & 6 \cdot L_2 & 2 \cdot L_2^{\,2} & 0 & -6 \cdot L_2 & 4 \cdot L_2^{\,2} \end{bmatrix}$$

$$k_2 = \begin{bmatrix} 1.185 \cdot 10^5 & 0 & 0 & -1.185 \cdot 10^5 & 0 & 0 \\ 0 & 104.063 & 416.25 & 0 & -104.063 & 416.25 \\ 0 & 416.25 & 2.22 \cdot 10^3 & 0 & -416.25 & 1.11 \cdot 10^3 \\ -1.185 \cdot 10^5 & 0 & 0 & 1.185 \cdot 10^5 & 0 & 0 \\ 0 & -104.063 & -416.25 & 0 & 104.063 & -416.25 \\ 0 & 416.25 & 1.11 \cdot 10^3 & 0 & -416.25 & 2.22 \cdot 10^3 \end{bmatrix}$$

$$c_2 := \frac{x_3 - x_2}{L_2} = 1 \qquad s_2 := \frac{y_3 - y_2}{L_2} = 0$$

$$T_2 := \begin{bmatrix} c_2 & s_2 & 0 & 0 & 0 & 0 \\ -s_2 & c_2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c_2 & s_2 & 0 \\ 0 & 0 & 0 & -s_2 & c_2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$K_2 := T_2^{\,\mathrm{T}} \cdot k_2 \cdot T_2 = \begin{bmatrix} 1.185 \cdot 10^5 & 0 & 0 & -1.185 \cdot 10^5 & 0 & 0 \\ 0 & 104.063 & 416.25 & 0 & -104.063 & 416.25 \\ 0 & 416.25 & 2.22 \cdot 10^3 & 0 & -416.25 & 1.11 \cdot 10^3 \\ -1.185 \cdot 10^5 & 0 & 0 & 1.185 \cdot 10^5 & 0 & 0 \\ 0 & -104.063 & -416.25 & 0 & 104.063 & -416.25 \\ 0 & 416.25 & 1.11 \cdot 10^3 & 0 & -416.25 & 2.22 \cdot 10^3 \end{bmatrix}$$

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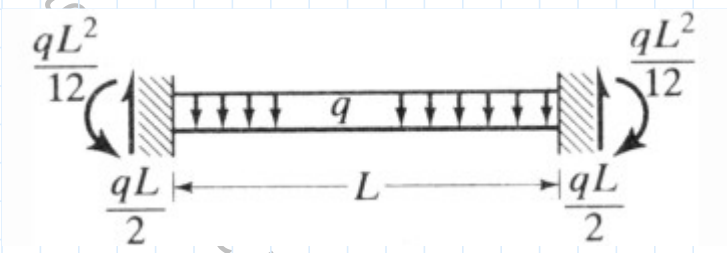
Fixed End Forces:

Element One:



$q := 24$ kN/M

$$F_{f1} = \begin{bmatrix} 0 \\ \frac{qL}{2} \\ \frac{qL^2}{12} \\ 0 \\ \frac{qL}{2} \\ \frac{-qL^2}{12} \end{bmatrix} \qquad F_{f1} := \begin{bmatrix} 0 \\ \frac{q \cdot L_1}{2} \\ \frac{q \cdot L_1^{\,2}}{12} \\ 0 \\ \frac{q \cdot L_1}{2} \\ \frac{-q \cdot L_1^{\,2}}{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 120 \\ 200 \\ 0 \\ 120 \\ -200 \end{bmatrix}$$

Rotate to Global Axis

$$F_{f1} := \begin{bmatrix} 120 \\ 0 \\ 200 \\ 120 \\ 0 \\ -200 \end{bmatrix}$$

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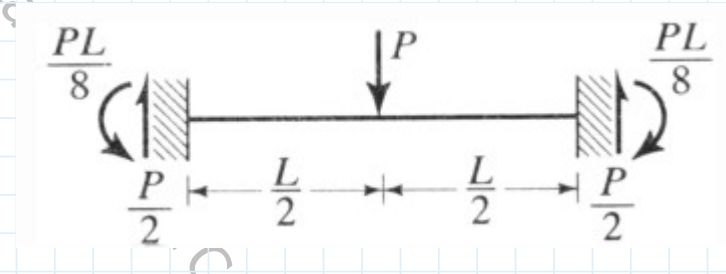
Element Two:



$P := -75$

$$F_{f2} = \begin{bmatrix} 0 \\ \frac{P}{2} \\ \frac{PL}{8} \\ 0 \\ \frac{P}{2} \\ \frac{-PL}{8} \end{bmatrix} \qquad F_{f2} := \begin{bmatrix} 0 \\ \frac{P}{2} \\ \frac{P \cdot L_2}{8} \\ 0 \\ \frac{P}{2} \\ \frac{-P \cdot L_2}{8} \end{bmatrix} = \begin{bmatrix} 0 \\ -37.5 \\ -75 \\ 0 \\ -37.5 \\ 75 \end{bmatrix}$$

Combine:

$F_{f1x1} := 120$ $\qquad F_{f2x2} := 0$

$F_{f1y1} := 0$ $\qquad F_{f2y2} := -37.5$

$F_{f1\theta1} := 200$ $\qquad F_{f2\theta2} := -75$

$F_{f1x2} := 120$ $\qquad F_{f2x3} := 0$

$F_{f1y2} := 0$ $\qquad F_{f2y3} := -37.5$

$F_{f1\theta2} := -200$ $\qquad F_{f2\theta3} := 75$

$$F_f := \begin{bmatrix} F_{f1x1} \\ F_{f1y1} \\ F_{f1\theta1} \\ F_{f1x2} + F_{f2x2} \\ F_{f1y2} + F_{f2y2} \\ F_{f1\theta2} + F_{f2\theta2} \\ F_{f2x3} \\ F_{f2y3} \\ F_{f2\theta3} \end{bmatrix} = \begin{bmatrix} 120 \\ 0 \\ 200 \\ 120 \\ -37.5 \\ -275 \\ 0 \\ -37.5 \\ 75 \end{bmatrix}$$

Created with Mathcad Express. See www.mathcad.com for more information.

Global Stiffness Matrix:

$$K_1 = \begin{bmatrix} 53.28 & 0 & -266.4 & -53.28 & 0 & -266.4 \\ 0 & 94800 & 0 & 0 & -94800 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 \\ -53.28 & 0 & 266.4 & 53.28 & 0 & 266.4 \\ 0 & -94800 & 0 & 0 & 94800 & 0 \\ -266.4 & 0 & 888 & 266.4 & 0 & 1776 \end{bmatrix}$$

$$K_2 = \begin{bmatrix} 118500 & 0 & 0 & -118500 & 0 & 0 \\ 0 & 104.063 & 416.25 & 0 & -104.063 & 416.25 \\ 0 & 416.25 & 2220 & 0 & -416.25 & 1110 \\ -118500 & 0 & 0 & 118500 & 0 & 0 \\ 0 & -104.063 & -416.25 & 0 & 104.063 & -416.25 \\ 0 & 416.25 & 1110 & 0 & -416.25 & 2220 \end{bmatrix}$$

$K_{1_11} := 53.28$ $K_{1_12} := 0$ $K_{1_13} := -266.4$ $K_{1_14} := -53.28$ $K_{1_15} := 0$ $K_{1_16} := -266.4$

$K_{1_21} := 0$ $K_{1_22} := 94800$ $K_{1_23} := 0$ $K_{1_24} := 0$ $K_{1_25} := -94800$ $K_{1_26} := 0$

$K_{1_31} := -266.4$ $K_{1_32} := 0$ $K_{1_33} := 1776$ $K_{1_34} := 266.4$ $K_{1_35} := 0$ $K_{1_36} := 888$

$K_{1_41} := -53.28$ $K_{1_42} := 0$ $K_{1_43} := 266.4$ $K_{1_44} := 53.28$ $K_{1_45} := 0$ $K_{1_46} := 266.4$

$K_{1_51} := 0$ $K_{1_52} := -94800$ $K_{1_53} := 0$ $K_{1_54} := 0$ $K_{1_55} := 94800$ $K_{1_56} := 0$

$K_{1_61} := -266.4$ $K_{1_62} := 0$ $K_{1_63} := 888$ $K_{1_64} := 266.4$ $K_{1_65} := 0$ $K_{1_66} := 1776$

$K_{2_44} := 118500$ $K_{2_45} := 0$ $K_{2_46} := 0$ $K_{2_47} := -118500$ $K_{2_48} := 0$ $K_{2_49} := 0$

$K_{2_54} := 0$ $K_{2_55} := 104.063$ $K_{2_56} := 416.25$ $K_{2_57} := 0$ $K_{2_58} := -104.063$ $K_{2_59} := 416.25$

$K_{2_64} := 0$ $K_{2_65} := 416.25$ $K_{2_66} := 2220$ $K_{2_67} := 0$ $K_{2_68} := -416.25$ $K_{2_69} := 1110$

$K_{2_74} := -118500$ $K_{2_75} := 0$ $K_{2_76} := 0$ $K_{2_77} := 118500$ $K_{2_78} := 0$ $K_{2_79} := 0$

$K_{2_84} := 0$ $K_{2_85} := -104.063$ $K_{2_86} := -416.25$ $K_{2_87} := 0$ $K_{2_88} := 104.063$ $K_{2_89} := -416.25$

$K_{2_94} := 0$ $K_{2_95} := 416.25$ $K_{2_96} := 1110$ $K_{2_97} := 0$ $K_{2_98} := -416.25$ $K_{2_99} := 2220$

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$$K := \begin{bmatrix} K_{1_11} & K_{1_12} & K_{1_13} & K_{1_14} & K_{1_15} & K_{1_16} & 0 & 0 & 0 \\ K_{1_21} & K_{1_22} & K_{1_23} & K_{1_24} & K_{1_25} & K_{1_26} & 0 & 0 & 0 \\ K_{1_31} & K_{1_32} & K_{1_33} & K_{1_34} & K_{1_35} & K_{1_36} & 0 & 0 & 0 \\ K_{1_41} & K_{1_42} & K_{1_43} & K_{1_44}+K_{2_44} & K_{1_45}+K_{2_45} & K_{1_46}+K_{2_46} & K_{2_47} & K_{2_48} & K_{2_49} \\ K_{1_51} & K_{1_52} & K_{1_53} & K_{1_54}+K_{2_54} & K_{1_55}+K_{2_55} & K_{1_56}+K_{2_56} & K_{2_57} & K_{2_58} & K_{2_59} \\ K_{1_61} & K_{1_62} & K_{1_63} & K_{1_64}+K_{2_64} & K_{1_65}+K_{2_65} & K_{1_66}+K_{2_66} & K_{2_67} & K_{2_68} & K_{2_69} \\ 0 & 0 & 0 & K_{2_74} & K_{2_75} & K_{2_76} & K_{2_77} & K_{2_78} & K_{2_79} \\ 0 & 0 & 0 & K_{2_84} & K_{2_85} & K_{2_86} & K_{2_87} & K_{2_88} & K_{2_89} \\ 0 & 0 & 0 & K_{2_94} & K_{2_95} & K_{2_96} & K_{2_97} & K_{2_98} & K_{2_99} \end{bmatrix}$$

$$K = \begin{bmatrix} 53.28 & 0 & -266.4 & -53.28 & 0 & -266.4 & 0 & 0 & 0 \\ 0 & 94800 & 0 & 0 & -94800 & 0 & 0 & 0 & 0 \\ -266.4 & 0 & 1776 & 266.4 & 0 & 888 & 0 & 0 & 0 \\ -53.28 & 0 & 266.4 & 118553.28 & 0 & 266.4 & -118500 & 0 & 0 \\ 0 & -94800 & 0 & 0 & 94904.063 & 416.25 & 0 & -104.063 & 416.25 \\ -266.4 & 0 & 888 & 266.4 & 416.25 & 3996 & 0 & -416.25 & 1110 \\ 0 & 0 & 0 & -118500 & 0 & 0 & 118500 & 0 & 0 \\ 0 & 0 & 0 & 0 & -104.063 & -416.25 & 0 & 104.063 & -416.25 \\ 0 & 0 & 0 & 0 & 416.25 & 1110 & 0 & -416.25 & 2220 \end{bmatrix}$$

Free DOFs are at 4, 5, 6, and 9

$$K_{ff} := \begin{bmatrix} 118553.28 & 0 & 266.4 & 0 \\ 0 & 94904.063 & 416.25 & 416.25 \\ 266.4 & 416.25 & 3996 & 1110 \\ 0 & 416.25 & 1110 & 2220 \end{bmatrix} \qquad F_{ff} := \begin{bmatrix} 120 \\ -37.5 \\ -275 \\ 75 \end{bmatrix}$$

$KU = F - F_f$ --> $U = K^{-1}\left(F - F_f\right)$ $F = 0$

$$U_f := K_{ff}^{-1} \cdot \left(-F_{ff}\right) = \begin{bmatrix} -0.001216 \\ 0.000344 \\ 0.09089 \\ -0.079293 \end{bmatrix} \begin{matrix} \text{m} \\ \text{m} \\ \text{radians} \\ \text{radians} \end{matrix}$$

$$U := \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.001216 \\ 0.000344 \\ 0.09089 \\ 0 \\ 0 \\ -0.079293 \end{bmatrix} \qquad u_1 := \begin{bmatrix} 0 \\ 0 \\ 0 \\ -0.001216 \\ 0.000344 \\ -0.09089 \end{bmatrix} \qquad u_2 := \begin{bmatrix} -0.001216 \\ 0.000344 \\ -0.09089 \\ 0 \\ 0 \\ -0.079293 \end{bmatrix}$$

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Local Element Forces:

$$f_e = k_e \cdot u_e + f_f$$

Element One:

$$f_1 := k_1 \cdot T_1 \cdot u_1 + T_1 \cdot F_{f1} = \begin{bmatrix} -32.611 \\ -144.278 \\ 118.966 \\ 32.611 \\ -95.722 \\ -361.745 \end{bmatrix}$$

Element Two:

$$f_2 := k_2 \cdot T_2 \cdot u_2 + F_{f2} = \begin{bmatrix} -144.096 \\ -108.303 \\ -364.648 \\ 144.096 \\ 33.303 \\ -201.775 \end{bmatrix}$$

Reactions:

$$K_{rf} := \begin{bmatrix} -53.28 & 0 & -266.4 & 0 \\ 0 & -9.48 \cdot 10^4 & 0 & 0 \\ 266.4 & 0 & 888 & 0 \\ -1.185 \cdot 10^5 & 0 & 0 & 0 \\ 0 & -104.063 & -416.25 & -416.25 \end{bmatrix}$$

$$F_{rf} := \begin{bmatrix} 120 \\ 0 \\ 200 \\ 0 \\ -37.5 \end{bmatrix} \qquad F_f = \begin{bmatrix} 120 \\ 0 \\ 200 \\ 120 \\ -37.5 \\ -275 \\ 0 \\ -37.5 \\ 75 \end{bmatrix}$$

$$R := K_{rf} \cdot U_f + F_{rf} = \begin{bmatrix} 126.815 \\ -56.831 \\ -222.802 \\ 113.185 \\ -18.169 \end{bmatrix}$$

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$u_{x1} := 126.815$

$u_{y1} := -56.831$

$\theta_1 := -222.802$

$u_{x3} := 113.185$

$u_{y3} := -18.169$

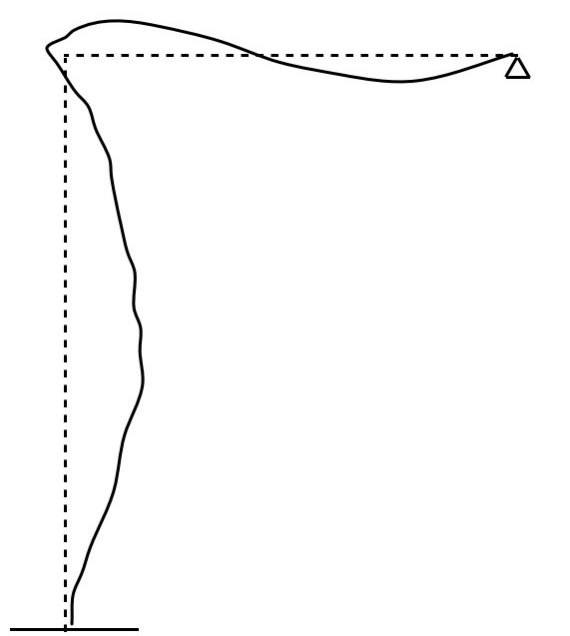
Global Equilibrium Check

$Forces_y := P = -75$ $\qquad$ $u_{y1} + u_{y3} - Forces_y = 0$

$Forces_x := q \cdot L_1 = 240$ $\qquad$ $u_{x1} + u_{x3} + Forces_x = 480$

$$\theta_1 - u_{x3} \cdot L_1 + u_{y3} \cdot (L_2) - P \cdot \left(\frac{L_2}{2}\right) - q \cdot L_1 \cdot \left(\frac{L_1}{2}\right) = -2400$$

Deformed Shape



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